

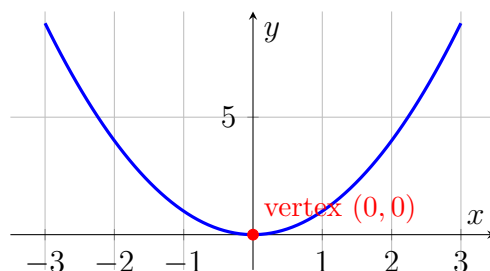
Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____ 11. _____

Identify the family, domain, and range

Example (Identify the family, domain, and range):

$f(x) = \frac{1}{x}$ is the **reciprocal** parent. A denominator can't be 0, so the domain is $x \neq 0$; the graph never touches $y = 0$ either, so the range is $y \neq 0$.



$y = x^2$ — the quadratic parent function.

Directions: Name the parent family, then give the domain and range.

1. $f(x) = x$

4. $f(x) = |x|$

2. $f(x) = x^2$

5. $f(x) = \sqrt{x}$

3. $f(x) = x^3$

6. $f(x) = \frac{1}{x}$

Match a shifted equation back to its parent family

Example (Match a shifted equation back to its parent family):

$g(x) = \sqrt{x} + 4$ still has a radical, so its parent family is **square root** (the +4 just shifts it later).

Directions: Even though these are shifted, name the **ORIGINAL** parent family only.

7. $g(x) = (x - 2)^2$

9. $g(x) = \frac{1}{x - 5}$

8. $g(x) = |x + 1| - 3$

10. (challenge) $g(x) = (x + 3)^3 + 1$

Challenge (same sheet):

11. Sketch all 6 parent functions from memory on one set of axes, labeling each with its equation.

Answer Key — fully worked

Procedure: Look for a root symbol, absolute value bars, or an exponent to identify the family.; Ignore any added/subtracted numbers outside — they only shift the graph, not the family.; Check for a denominator or a radical: these create domain restrictions.; State domain first, then determine range from the graph's lowest or highest point..

1. Linear; domain: all real numbers; range: all real numbers
2. Quadratic; domain: all real numbers; range: $y \geq 0$
3. Cubic; domain: all real numbers; range: all real numbers
4. Absolute value; domain: all real numbers; range: $y \geq 0$
5. Square root; domain: $x \geq 0$; range: $y \geq 0$
6. Reciprocal; domain: $x \neq 0$; range: $y \neq 0$
7. Quadratic (the $(x - 2)^2$ is still squared, just shifted right)
8. Absolute value (the bars are still there, just shifted)
9. Reciprocal (still a fraction with x in the denominator)
10. Cubic — the exponent of 3 on $(x + 3)$ shows this came from the cubic parent, just shifted left 3 and up 1
11. Six correct sketches on one grid: $y = x$ (straight line through origin, slope 1); $y = x^2$ (U-shaped parabola, vertex $(0, 0)$); $y = x^3$ (S-shaped curve through origin, rising left to right); $y = |x|$ (V-shape, vertex $(0, 0)$, opening up); $y = \sqrt{x}$ (half-parabola starting at $(0, 0)$, in the first quadrant only); $y = \frac{1}{x}$ (two hyperbola branches in quadrants I and III, never touching either axis). Each curve labeled with its equation; $\sqrt{x}$ and $1/x$ should visibly show their restricted domains.